

Modul infAAI-01a: Agentic AI

Modulverantwortliche(r): Prof. Dr. Hauke Schramm					
Präsenzzeiten:	2V 2Ü	ECTS:	6	Turnus:	jedes Jahr
Workload:	30 h lectures, 30 h exercises, 120 h self studies				
Modulkategorien:	WO_approval (Sonstige)				
Dauer:	ein Semester	Lehrsprache:	Englisch		
Kurzfassung: This master's-level lecture addresses the design, implementation, and operation of autonomous Large Language Model (LLM) based agent systems. Starting from the modern agent loop and its classical predecessors, the course develops the fundamental mechanisms of LLM-based agents, like function calling, tool integration via the Model Context Protocol (MCP), prompt engineering, reasoning models, and the principal agentic design patterns. Higher-level aspects such as orchestration, memory and Retrieval-Augmented Generation (RAG), observability, multi-agent collaboration, and self-reflection are subsequently treated as architectural choices with explicit cost, latency, and security trade-offs. The course concludes with agent security, the evaluation of agentic systems, and the data-protection and accountability implications of autonomous behaviour. It is intended for students familiar with the fundamentals of machine learning and LLMs but new to LLM-based agentic systems. Throughout the semester, students apply the lecture concepts in practical lab work as a series of focused experiments on individual lecture topics.					
Kompetenzziele: Students are able to • explain the agent loop and apply it to a concrete problem, • select agentic design patterns appropriate to a given problem and delineate them from classical agent architectures, • choose and justify architectural decisions for LLM-agent infrastructure along functionality, cost, latency, and security, • design, implement, instrument, and evaluate LLM agents, including tool integration via function calling and the MCP, • reflect on how reasoning models reshape agentic system design, including chain-of-thought redundancy and cost/latency trade-offs, • decide when to use a framework (e.g. LangChain/LangGraph) and when a lean plain-code implementation is preferable, • design multi-agent architectures and justify their use relative to single-agent alternatives, • explain self-reflection as a self-improvement mechanism and reflect on its risks, • recognise security risks and design countermeasures, and • evaluate agentic systems and reflect on data protection, accountability, and the energy and cost footprint.					

Lehrinhalte:

- Introduction to Agentic AI, demarcation from workflows and "agent-washing"
- Classical agent architectures as historical predecessors of the LLM agent loop
- The agent loop: LLM, tool calls, observation, context buffer, stop conditions
- Overview of agentic design patterns: planning, tool use, reflection, multi-agent
- Function calling as the basic mechanism for tool use
- Tool integration via the Model Context Protocol (MCP)
- Prompt engineering for agents (system prompts, few-shot, chain-of-thought, output formatting)
- Reasoning models: internal reasoning vs. explicit chain-of-thought, implications for agent design
- Orchestration: Directed Acyclic Graph (DAG) based vs. dynamic, ReAct (Reasoning and Acting), plan-and-execute
- Framework-based vs. plain-code agents
- Memory infrastructure, vector databases, and retrieval-augmented generation (RAG)
- Sandboxing, guardrails, observability, and debugging strategies for non-deterministic agent runs
- Multi-agent collaboration: architectural patterns, role specialisation, frameworks, scaling limits and anti-patterns
- Self-reflection as a self-improvement mechanism (Reflexion framework)
- Agent security: prompt injection (direct and indirect), confused deputy, tool misuse, countermeasures
- Evaluation of agentic systems: LLM-as-judge, trajectory evaluation, selected benchmarks
- Data protection, accountability, energy and cost footprint of autonomous agents

Voraussetzungen:

- Foundations of machine learning and data science
- Foundations of deep learning and transformers
- Basic understanding of LLMs (tokenisation, prompting, context windows)
- Solid programming skills in Python
- Reading, writing and listening competence in academic English

Prüfungsleistung: Written exam at the end of the semester. Successful participation in the exercises (homework assignments) is a prerequisite to participate in the exam.

Lehr- und Lernmethoden: Lecture and lab are tightly coupled, with the lab demanding active engagement from the students. The teaching is organised around the following components:

- Structured lecture based on slide decks
- Board work with illustrative examples
- Flipped classroom elements based on pre-recorded videos
- Design and code analysis of selected agents
- Self-assessment quizzes with sample solutions and lecture-accompanying question catalogues
- Weekly worksheets containing conceptual questions and short design exercises
- Implementation tasks in Python, building incrementally on previous topics

Verwendbarkeit: This course is applicable for students pursuing computer science, data science, business informatics and related fields.

Literatur:

- Russell, S.; Norvig, P. (2020): Artificial Intelligence: A Modern Approach. 4th edition. Pearson. (Chapters on intelligent agents)
- Yao, S.; Zhao, J.; Yu, D.; Du, N.; Shafran, I.; Narasimhan, K.; Cao, Y. (2023): ReAct: Synergizing Reasoning and Acting in Language Models. International Conference on Learning Representations (ICLR 2023). arXiv:2210.03629.
- Shinn, N.; Cassano, F.; Berman, E.; Gopinath, A.; Narasimhan, K.; Yao, S.: *Reflexion: language agents with verbal reinforcement learning.* In Advances in Neural Information Processing Systems, Bd. 36, Curran Associates, 2023.
- Greshake, K.; Abdelnabi, S.; Mishra, S.; Endres, C.; Holz, T.; Fritz, M. (2023): Not what you've signed up for: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection. 16th ACM Workshop on Artificial Intelligence and Security (AISec '23), co-located with ACM CCS 2023. arXiv:2302.12173.
- Schluntz, E.; Zhang, B. (2024): Building Effective Agents. Anthropic Engineering Blog, December 20, 2024.
- Wang, L. et al. (2024): A Survey on Large Language Model based Autonomous Agents. Frontiers of Computer Science 18(6). DOI: 10.1007/s11704-024-40231-1.

Further Resources

- Ng, A. (2024): Agentic AI. Online course, DeepLearning.AI.
- Song, D. et al. (2024-2025): Large Language Model Agents and Advanced Large Language Model Agents. MOOC series, University of California, Berkeley.

Verweise: **Agentic AI** connects with several modules in the local portfolio to form coherent learning paths:

- **Foundations of Deep Learning** (infFDL-01a), **Generative AI** (infGAI-01a) and **Deep Learning for Natural Language Processing** (infDLfNLP-01a) give students the transformer and language-model foundations that make the technical content of **Agentic AI** fully accessible.
- **LLM-based System Engineering** (infLLMSe-01a) builds practical proficiency in applying LLMs in software engineering and business-process modelling, which students can then elevate to full agent architectures in **Agentic AI**.
- **Bachelor project - Generative AI Agents in Knowledge-Based Work** (infGKIW-01a) and **Collaboration Engineering** (infCE-01a) develop the socio-technical competence to identify *which* agents to build and *how* to embed them in human work. **Agentic AI** builds on this groundwork, providing the technical-architectural depth needed to actually design, implement, secure, and evaluate such agents.
- **Autonomous Learning** (infAuLearn-01a) deepens the understanding of how systems *learn* to behave autonomously (reinforcement learning, active learning) a perspective that broadens the agentic-AI view by combining learned and orchestrated forms of autonomy.

Kommentar: